

14 Which statements about the Michaelis–Menten constant (K_m) of an enzyme are correct?

- 1 At the K_m value, half the active sites of the enzyme should be occupied by the substrate.
- 2 K_m represents the substrate concentration at which the enzyme is working at half its maximum rate.
- 3 The lower the K_m value, the lower the affinity of the enzyme to its substrate.
- 4 When an enzyme has a high K_m value, the enzyme-catalysed reaction will proceed very slowly to its maximum rate.

A 1, 2 and 3 **B** 1, 2 and 4 **C** 1 and 3 only **D** 2 and 4 only